AI ENGINEERING JOURNAL

Team name:	PYRAMID
AI Tools used:	Claude via Claude Code for implementation, testing, debugging and UI refinement. Within the MVP, Gemini or Claude can support order intake and product-data suggestions; automated tests use a deterministic mock.
Language:	English

1. AI Strategy

Briefly describe how your team used AI during the project.

(Max. 150 words).

PYRAMID used AI as an engineering accelerator, not a decision-maker. The team interpreted the Round 2 brief and Round 1 proposal, then set the scope, architecture, data model, acceptance criteria and daily plan. Claude, via Claude Code, helped implement clearly defined modules, draft tests, diagnose failures and refine the interface against a team-selected direction. We treated every output as a proposal: a team member reviewed the code and screens, challenged anything that conflicted with our design rules, and required unit, integration, browser and Linux CI checks before acceptance. The same principle shapes the MVP. AI converts unstructured order text into reviewable draft lines and suggests product-data mappings; people approve those suggestions, while stock availability, routing, reservations and ETA calculations remain deterministic.

2. AI Usage log

Document the most significant AI interactions during your project. Maximum 5 tasks

Stage	Purpose of using AI	Original time estimate	Actual time with AI	Human Validation	Outcome
Build: core services	Implement the Sweden legacy simulator and Local SE services—inventory sync, rules-based routing, reservations and the order state machine—from team-defined checkpoints and acceptance tests T01–T17	≈2 days (plan: 4–5 Nov)	≈1 day (4–5 Nov)	Reviewed every checkpoint against the specification. We identified five issues—including row-age freshness, post-approval reservation expiry, an LLM call inside a database transaction and missing endpoints—and required a regression test for each fix	Core backend completed; 294 automated tests and a 30-check smoke suite passed
Build: journeys and integration	Implement the customer, staff and admin journeys, the Central dashboard and service integration from the team's user journey and API design	≈2 days (plan: 5–6 Nov)	≈0.5 day (5 Nov)	Walked the golden journey in a real browser, validated a live Gemini call end to end, exercised outage paths and scanned logs for secrets (41/41 checks passed)	Golden journey completed end to end
Validate: browser tests and hardening	Create Playwright coverage for team-defined scenarios: split fulfilment, shortage, AI review, outages and authorisation	≈1 day (plan: 7 Nov)	≈0.5 day (5 Nov)	Defined expected outcomes before execution, reviewed every failure and rejected a 98% AI-accuracy claim because it came from a deterministic mock	27 browser scenarios passed; limitations and unvalidated claims documented
Round 2: close Round 1 gaps	Implement the four gaps identified against our Round 1 commitments: pilot KPI card, per-store circuit breaker, accept-available path and AI order reader	≈1.5 days (plan: 7 Nov)	≈4 hours (5 Nov)	Mapped every change back to the Round 1 slides and added 43 tests. The full browser suite exposed a bug in the mock order reader, which we fixed with a regression test	All four gaps closed; 492 unit/integration tests and 29/29 browser scenarios passed; Linux CI green
Polish: interface refresh	Refresh the web interface in a team-selected, Stripe/Linear-inspired direction without changing behaviour	Not in plan (≈1 day by hand)	≈2 hours (5 Nov)	Selected the direction from three mock-ups and reviewed desktop and mobile captures. Disabled a font feature that changed 900x2400 to 900×2400 and reworked the mobile order-step bar	Refreshed UI; 80/80 web tests, the browser suite and Linux CI passed

3. Key AI Decisions

Describe **2–3 examples** where AI had a meaningful impact on your project.

Task	AI suggestion	What your team did
Inventory trust: when is stock reliable enough to promise?	Judge freshness from the age of each stock row	Rejected. Unchanged inventory is not stale. Trust now depends on the latest verified change-feed check, cursor continuity and source connectivity, with regression coverage.
AI normalisation worker	Keep a database transaction open while waiting for the LLM	Changed to a committed work claim with a lease, no open transaction during the AI call, and limited retries before human takeover. A slow provider can no longer block order processing.

Reporting AI quality	Report 98% accuracy for product-data suggestions	Rejected. The figure came from a deterministic mock, not a real model. We state that real-model accuracy is unmeasured, and every suggestion still requires human approval.

4. Reflection

Answer each question in 2-4 sentences

1. Where did AI help your team the most?

   AI contributed most by translating our written specifications into working code and executable tests. This shortened the initial implementation and left more time for review, exception handling and closing gaps against our Round 1 commitments. It also accelerated debugging; for example, it helped trace a failed browser scenario to the mock order reader's sentence-splitting logic.

2. What AI output did you reject or modify, and why?
   We rejected outputs that looked plausible but violated core design rules: using row age as stock freshness, waiting for an LLM inside a database transaction, and allowing approved reservations to expire. Each correction was backed by a regression test. We also rejected a 98% accuracy claim produced by the deterministic mock and corrected a font setting that changed "900x2400" to "900×2400", because both evidence and customer input must be represented faithfully.

3. What engineering or business decisions required human judgment?
   Human judgment set the pilot scope: one Swedish region, the Round 1 success measures (OTIF +3 pp, status calls −30%, adoption >90%) and the commitments to prove first. The team also decided that AI may interpret text but must never determine stock, routing, reservations or ETAs, and that every suggestion requires approval. Business trade-offs—including the 10% safety buffer, the five-minute node timeout and how simulated results are labelled—also remained team decisions.

4. Are there any obstacles when using AI?
   AI output can be confidently wrong, so every contribution required review and test evidence; that assurance work takes real time. The live Gemini API was also intermittent in our final check, with two of three calls returning HTTP 503, so the MVP includes a fallback and the automated tests use a deterministic mock. We therefore make no claim about real-model accuracy and use written checkpoints to keep AI-assisted work focused and reviewable.